KARTHIKEYAN P

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PROFESSIONAL SUMMARY

Full Stack Developer (MERN) with three internships and hands-on experience building secure, role-based web applications with REST APIs. Currently building an AI-powered donation platform using computer vision, OCR and geospatial matching.

SKILLS

- **Languages:** Python, Java, C, JavaScript (ES6+)
- **Frontend:** React.js, React Native, HTML5, CSS3
- **Backend & Databases:** Node.js, Express.js, FastAPI, REST APIs, JWT, MongoDB (Atlas, Geospatial), MySQL
- **AI / Computer Vision:** YOLOv8/v11, OpenCV, OCR, Google Gemini API, MongoDB Vector Search & RAG
- **Tools & Cloud:** Git, GitHub, Azure Fundamentals, AWS, OSRM, Gradio

WORK EXPERIENCE

Web Developer | *K.L.N. Innovation & Research Park (KLN IRP), Madurai* **Dec 2024 – Present**

- Own end-to-end web development of an in-house research product with faculty mentors.
- Design and test application workflows, improving code quality, execution speed and product stability.

MERN Stack Development Intern | *Elysium Technologies, Madurai* **Jun 2026**

- Built production-style full-stack components using React.js, Node.js, Express.js and MongoDB.

MERN Stack Development Intern | *Vinsup Technology, Madurai* **Dec 2024 – Jan 2025**

- Developed backend modules and responsive frontend interfaces; designed, tested and debugged REST API endpoints.

PROJECTS

HSDE — Hyperlocal Smart Donation Engine | *Final-Year Project* **Training & Testing Phase**

- Mobile app (**React Native**, **FastAPI**, **MongoDB Atlas**) connecting NGOs, donors and volunteers across 10 categories.
- Built a matching engine to find donors within a **15 km radius**, ranked by urgency and category; added **YOLO + OCR** verification and **OSRM** volunteer routing.

E-MBook — Digital Measurement Book | *Nimirndhu Nil Hackathon '26* **Jun 2026**

- **MERN** app digitizing the paper Measurement Book for the Rural Development & Panchayat Raj Department.
- Built role-based access and a **6-tier approval workflow** (JE → AE → AEE → EE → SE → CE) with dashboards.

Hospireo — Healthcare Discovery Platform | *Full Stack Engineer* **Testing Phase**

- Hospital and doctor discovery platform with **Node.js/Express.js** APIs, department-wise doctor profiles and real-time availability on **MongoDB**.

AI Resume PDF Generator | *Independent Project* **Aug 2025**

- Used the **Gemini API** to turn raw input into structured resume text, with a **Gradio** interface and **ReportLab** PDF export.

EDUCATION

B.E., Computer Science and Engineering | *K.L.N. College of Engineering* | *CGPA: 8.71 / 10* **Expected May 2027**

Higher Secondary Course (HSC) | *A.P.T. Dorairaj Higher Secondary School* | *80.83%* **2023**

CERTIFICATIONS

- **Cloud & AI:** Azure Fundamentals (Microsoft); AWS Security & Serverless; MongoDB – Vector Search, RAG, AI Agents
- **Others:** PostgreSQL, PHP & MySQL (IIT Bombay); Cloud Computing, Ethical Hacking (NPTEL); MERN Stack (Vinsup); Generative AI Bootcamp (NoviTech)

ACHIEVEMENTS & LEADERSHIP

- Level 2 Ideation Camp, Nimirndhu Nil Hackathon '26; Smart India Hackathon participant; Presenter, KLNICST '26.
- Technical Lead, Technozare Symposium; Technical Core Member, GDSC; Outreach Lead, Zeigen '26.